

LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
Experiment No.	6
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1. TITLE

JavaScript Functions and Classes – Creating Multiple Objects Using a Class

2. AIM

To differentiate between JavaScript functions and classes, and to demonstrate how a class can be used to create multiple objects with common properties and methods.

3. OBJECTIVE

- To understand the difference between functions and classes in JavaScript.
 - To understand classes as blueprints for creating objects.
 - To define properties using a constructor.
 - To define methods inside a class.
 - To create multiple objects using the new keyword.
 - To access object properties and methods.
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4. THEORY

A function in JavaScript is a block of code used to perform a particular operation. A class is a blueprint used to create objects with common properties and methods. Classes help group related data and operations together.

In this program, the Student class represents a common structure for student objects. The constructor initializes the name, roll number, and marks of each student. The showStudent() method displays the details. Multiple objects are created from the same class, but each object stores different student information.

DIFFERENCE BETWEEN FUNCTIONS AND CLASSES

Functions	Classes
Used to perform a particular operation.	Used as a blueprint for creating objects.
Called when its operation needs to be executed.	Objects are created using the new keyword.

Usually focuses on a specific task.	Groups related properties and methods.
Can work independently.	Defines a common structure for objects.
Example: calculating average marks.	Example: creating different Student objects.

5. ALGORITHM / PROCEDURE

1. Create a class named Student.
 2. Define name, roll number, and marks in the constructor.
 3. Create a showStudent() method to display student details.
 4. Create the first Student object using new.
 5. Create two more Student objects using the same class.
 6. Store different details in each object.
 7. Call showStudent() for each object.
 8. Display the student details in the console.
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6. PROGRAM

```
class Student {
    constructor(name, rollNo, marks) {
        this.name = name;
        this.rollNo = rollNo;
        this.marks = marks;
    }

    showStudent() {
        console.log(this.name + " - " + this.rollNo);
        console.log("Marks: " + this.marks);
    }
}

let s1 = new Student("Kavya", "A101", 85);
let s2 = new Student("Rahul", "A102", 91);
let s3 = new Student("Sneha", "A103", 88);

s1.showStudent();
s2.showStudent();
s3.showStudent();
```

7. CODE EXPLANATION

Code / Function	Explanation
class Student	Creates the Student class and defines a common

	structure.
constructor()	Initializes the values of each student object.
this.name	Stores the student's name.
this.rollNo	Stores the student's roll number.
this.marks	Stores the student's marks.
showStudent()	Displays the details of a student object.
new Student()	Creates a new object from the Student class.
s1, s2, s3	Three separate Student objects with different values.
s1.showStudent()	Calls the method for the first object.

8. OUTPUT

```

● PS D:\kavya\Tutorials\FS lab\Observation_Tasks\task 6\js_func_class> node code.js
Name: Ravi
Roll No: 101
Department: CSE
Name: Priya
Roll No: 102
Department: ECE
Name: Arun
Roll No: 103
Department: IT
❖ PS D:\kavya\Tutorials\FS lab\Observation_Tasks\task 6\js_func_class>

```

9. RESULT

Thus, the difference between JavaScript functions and classes was studied successfully. A Student class was created and multiple objects were generated from it with common properties and methods. Each object stored different student details and displayed them successfully.